Task 3 Report: Improvement, Ablation, Statistical Testing and Explainability

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1. Aim and selection protocol

Task 3 separates model selection from final evaluation. Ten candidate configurations are ranked on validation Macro-F1, while the held-out test set is opened only for the selected CBAM configuration and its matched plain control. Using the test set during selection would make it part of the tuning process and produce an optimistically biased estimate.

The exact-duplicate-group partition from Task 2 is retained. Every run permits at most 60 epochs and uses early stopping. For the final paired comparison, the plain network is obtained from the selected CBAM configuration by removing attention only; dropout, depth, batch normalization, augmentation, optimizer and learning-rate schedule remain fixed.

2. Ablation design

The ablation varies attention type, dropout, batch normalization, backbone depth, augmentation and learning-rate scheduling. Input resolution and kernel sizes remain fixed so that the run budget is concentrated on the components most directly connected to the proposed architecture. Configuration J, the plain CNN without dropout, completes the factorial attention-by-dropout control that was missing from the initial analysis.

ID	Configuration	Attention	Dropout	BN	Blocks	Aug.	LR	Val. Macro-F1	Recall	Epochs
A	Plain CNN	None	Yes	Yes	4	Yes	Plateau	0.9542	0.9303	27
B	Channel only	Channel	Yes	Yes	4	Yes	Plateau	0.9467	0.9581	21
C	Spatial only	Spatial	Yes	Yes	4	Yes	Plateau	0.9515	0.9320	9
D	Full CBAM	CBAM	Yes	Yes	4	Yes	Plateau	0.9351	0.8980	10
E	CBAM, no dropout	CBAM	No	Yes	4	Yes	Plateau	0.9714	0.9766	19
F	CBAM, no batch norm	CBAM	Yes	No	4	Yes	Plateau	0.9530	0.9461	16
G	CBAM, 3 blocks	CBAM	Yes	Yes	3	Yes	Plateau	0.9560	0.9251	42
H	CBAM, augmentation off	CBAM	Yes	Yes	4	No	Plateau	0.4514	0.0680	10
I	CBAM, cosine LR	CBAM	Yes	Yes	4	Yes	Cosine	0.9093	0.7950	10

ID	Configuration	Attention	Dropout	BN	Blocks	Aug.	LR	Val. Macro-F1	Recall	Epochs
J	Plain CNN, no dropout	None	No	Yes	4	Yes	Plateau	0.9600	0.9289	17

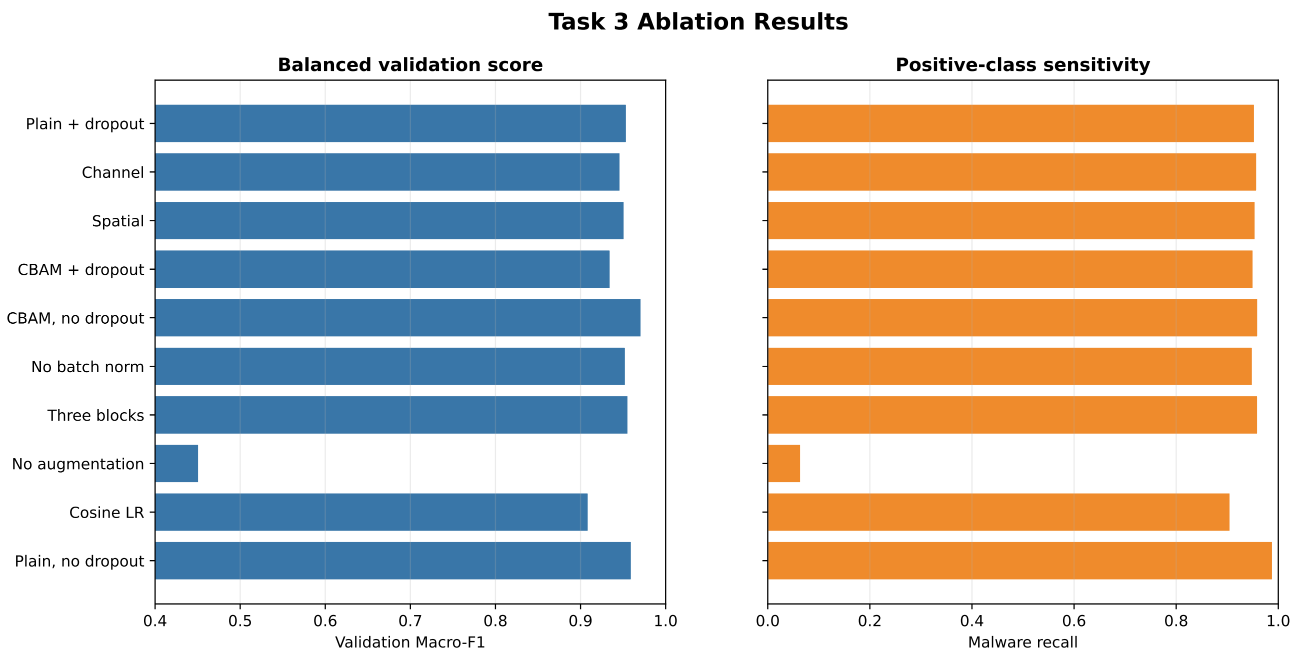

Figure 1: Validation Macro-F1 and malware recall for the ten ablation configurations.

Augmentation is the dominant ablation factor in the recorded validation run. Without it, configuration H falls to 0.4514 Macro-F1 and 0.0680 malware recall, indicating near-collapse of positive-class detection at threshold 0.5. The result establishes dependence on the training perturbation for this split; it does not identify which feature fields require the perturbation.

With dropout enabled, channel-only and spatial-only attention reach 0.9467 and 0.9515 Macro-F1, respectively, compared with 0.9542 for the plain CNN. Full CBAM drops to 0.9351. Removing batch normalization gives 0.9530, reducing the backbone to three blocks gives 0.9560, and cosine learning-rate scheduling gives 0.9093. No attention variant with dropout exceeds its matched plain control.

3. Attention and dropout control

Model	Attention	Dropout	Validation accuracy	Validation Macro-F1	ROC-AUC	PR-AUC
Plain CNN	No	Yes	0.9570	0.9542	0.9921	0.9810
Plain CNN	No	No	0.9626	0.9600	0.9931	0.9737
CNN+CBAM	Yes	Yes	0.9394	0.9351	0.9857	0.9678
CNN+CBAM	Yes	No	0.9729	0.9714	0.9956	0.9892

Table 2. Two-by-two validation control for attention and dropout.

The factorial control changes the interpretation of the validation results. With dropout enabled, CBAM reduces Macro-F1 by 0.0191; without dropout, CBAM increases it by 0.0114. Dropout reduces the plain model by 0.0058 and the CBAM model by 0.0363. Attention therefore cannot be credited for the validation gain without specifying the dropout condition.

CBAM without dropout is selected at 0.9714 validation Macro-F1. A plausible explanation is that dropout disrupts feature maps that the attention gates are trying to reweight, but the present experiment does not directly test that mechanism. The interaction is reported as an empirical result rather than a causal account.

4. Matched held-out comparison

Model	Accuracy	Malware precision	Malware recall	Malware F1	Macro-F1	Weighted F1	ROC-AUC	PR-AUC
Plain CNN, no dropout	0.9785	0.9532	0.9929	0.9727	0.9775	0.9786	0.9980	0.9968
Final CBAM, no dropout	0.9587	0.9794	0.9120	0.9445	0.9558	0.9584	0.9938	0.9885

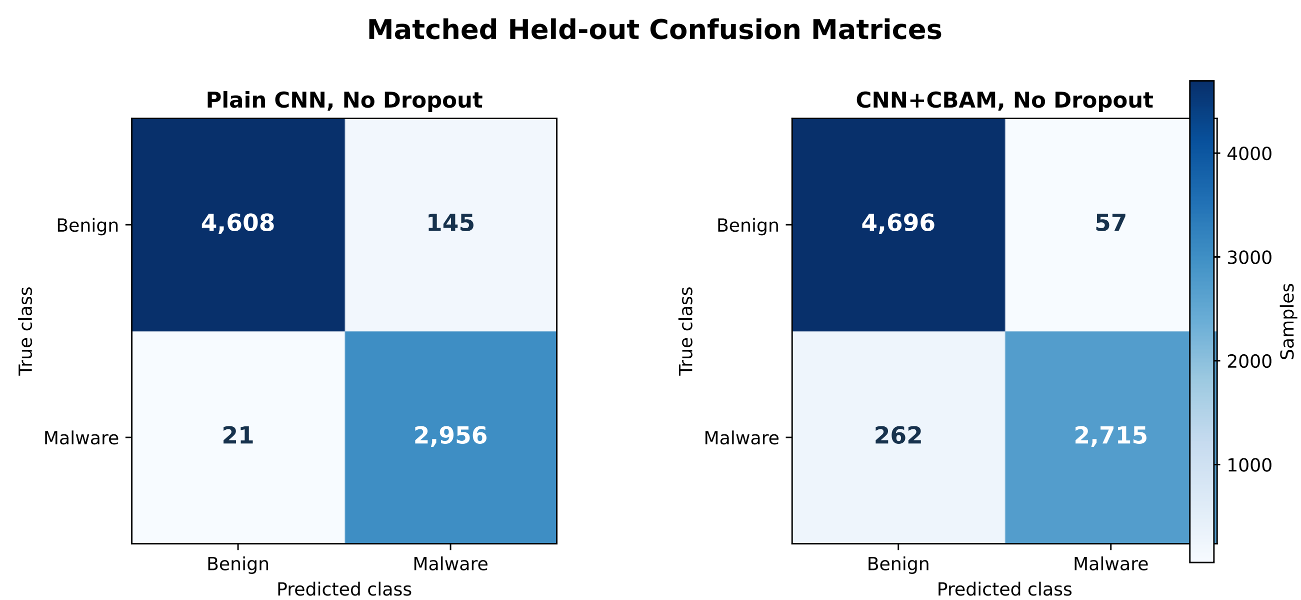

Figure 2: Confusion matrices for the matched plain CNN and final CBAM model.

On the untouched test set, the plain CNN without dropout reaches 0.9775 Macro-F1, whereas the selected CBAM model reaches 0.9558. CBAM improves malware precision from 0.9532 to 0.9794 but reduces recall from 0.9929 to 0.9120. The confusion matrices show the practical consequence: CBAM removes 88 benign false alarms but adds 241 malware misses. At threshold 0.5, that exchange is unfavorable when false negatives carry the higher security cost.

Ranking metrics agree with the threshold-specific result. The plain CNN records ROC-AUC 0.9980 and PR-AUC 0.9968, compared with 0.9938 and 0.9885 for CBAM. Thus the reversal from validation is not confined to the 0.5 decision boundary; the plain network also ranks the held-out samples more effectively overall.

5. McNemar test

McNemar's exact test compares the two decisions made on each of the 7,730 test samples. CBAM alone is correct for 122 discordant cases, whereas the plain CNN alone is correct for 275. The exact p-value is 1.11×10^{-14}, which

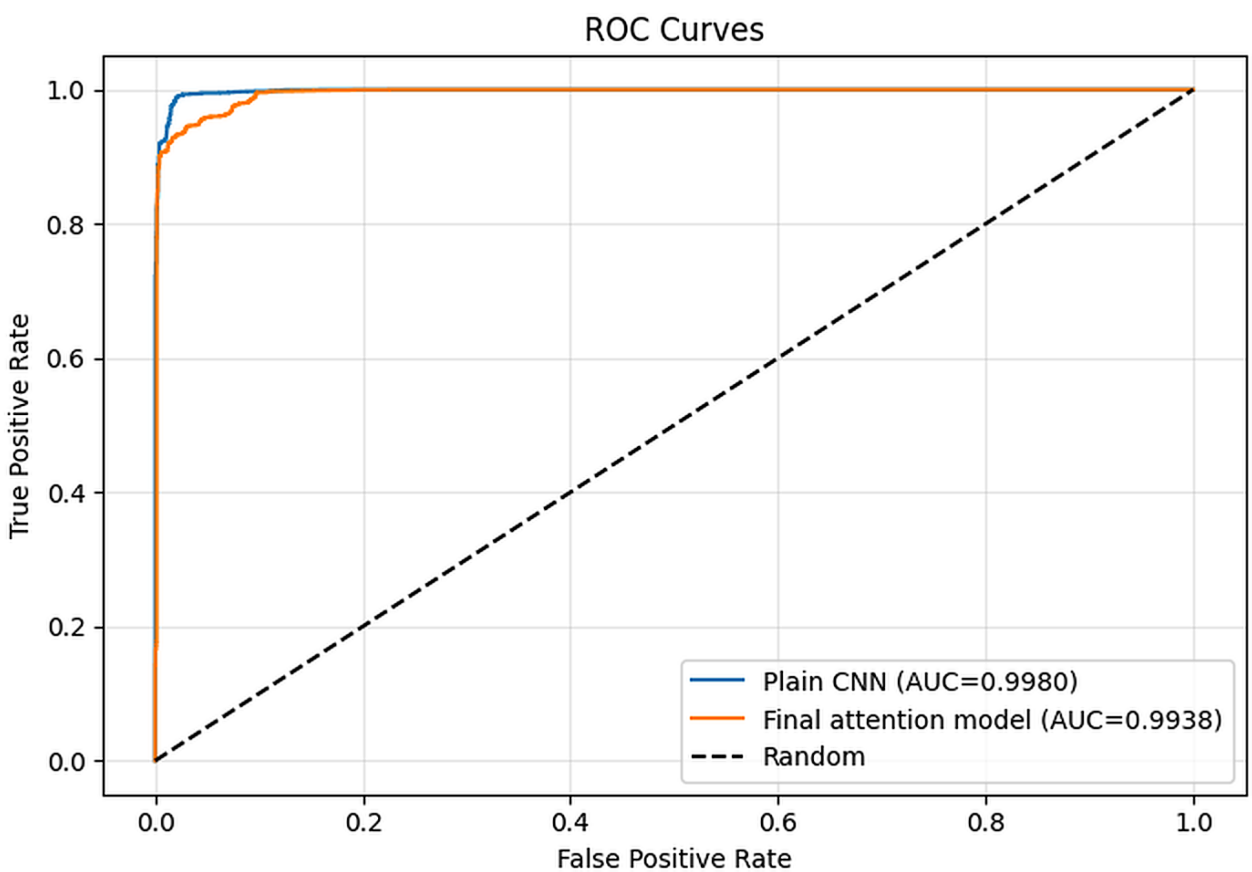

Figure 3: Held-out ROC curves for the matched models.

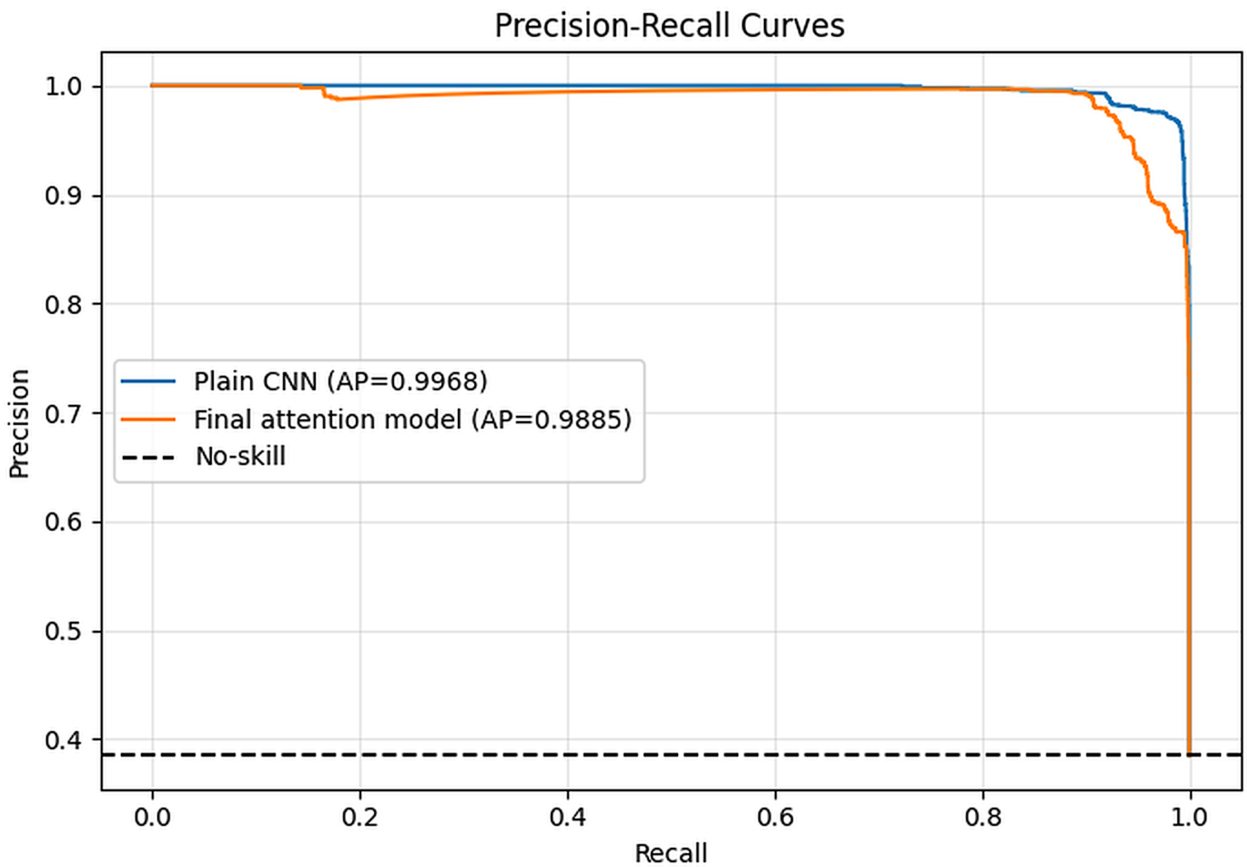

Figure 4: Held-out precision-recall curves for the matched models.

rejects equal paired error rates at $\alpha = 0.05$. The direction favors the plain network because substantially more discordant samples are correct only under that model.

This test applies only to models evaluated on the same samples. Published SOMLAP accuracies cannot enter a McNemar comparison because their per-sample predictions and test membership are unavailable.

6. Leakage-safe five-fold cross-validation

Five StratifiedGroupKFold partitions preserve approximate class balance while keeping every exact feature-vector group inside one fold. The plain CNN and CBAM use identical fold assignments, so their fold Macro-F1 values form paired observations rather than independent runs.

Fold	Plain CNN Macro-F1	CBAM Macro-F1	Plain minus CBAM
1	0.9829	0.9738	+0.0091
2	0.9864	0.9848	+0.0016
3	0.9834	0.9822	+0.0012
4	0.9829	0.9826	+0.0003
5	0.9756	0.9808	-0.0052
Mean ± SD	0.9823 ± 0.0040	0.9809 ± 0.0042	+0.0014

Table 4. Paired grouped five-fold Macro-F1 results.

The plain CNN is higher in four folds and CBAM in one. Mean Macro-F1 is 0.9823 ± 0.0040 for the plain model and 0.9809 ± 0.0042 for CBAM, a mean difference of 0.0014. The Wilcoxon signed-rank statistic is $W = 4.0$ with $p = 0.4375$; five fold pairs do not provide evidence of a population-level difference at $\alpha = 0.05$.

McNemar and Wilcoxon answer different questions. McNemar uses thousands of paired sample outcomes from one fixed test set and detects a directional error difference there. Wilcoxon uses five aggregate fold scores and has much lower power. Reporting both prevents the significant held-out result from being misrepresented as stable superiority across folds.

7. Context against published SOMLAP results

Result	Protocol	Accuracy	Weighted F1	Comparison status
Plain CNN (this study)	Exact-group held-out test	0.9785	0.9786	Paired under the current protocol
Final CBAM (this study)	Exact-group held-out test	0.9587	0.9584	Paired under the current protocol
Kattamuri et al. [2]	ACO with 12 selected features	0.9937	Not reported	Different features and split
Alwateer et al. [7]	Hierarchical deep network	0.9911	0.9885	No shared split or predictions

Table 5. SOMLAP context. Published rows are not protocol-matched paired comparisons.

Both published SOMLAP studies report higher accuracy than either current model. Kattamuri et al. use ACO to reduce the representation to 12 features [2], whereas Alwateer et al. use a hierarchical deep network [7]. Their split rules and predictions are unavailable, so the values establish numerical context but not a controlled ranking. Under the teacher's fairness rule, the current result remains below the strongest comparable published number.

8. Grad-CAM and LIME

The explanation analysis reconstructs the fixed grouped test partition and loads the CBAM model selected during ablation. Two cases are examined: a benign record correctly assigned a malware probability of 0.072, and a benign record incorrectly assigned a malware probability of 0.992. The second is therefore a high-confidence false positive rather than a borderline error.

8.1 Grad-CAM

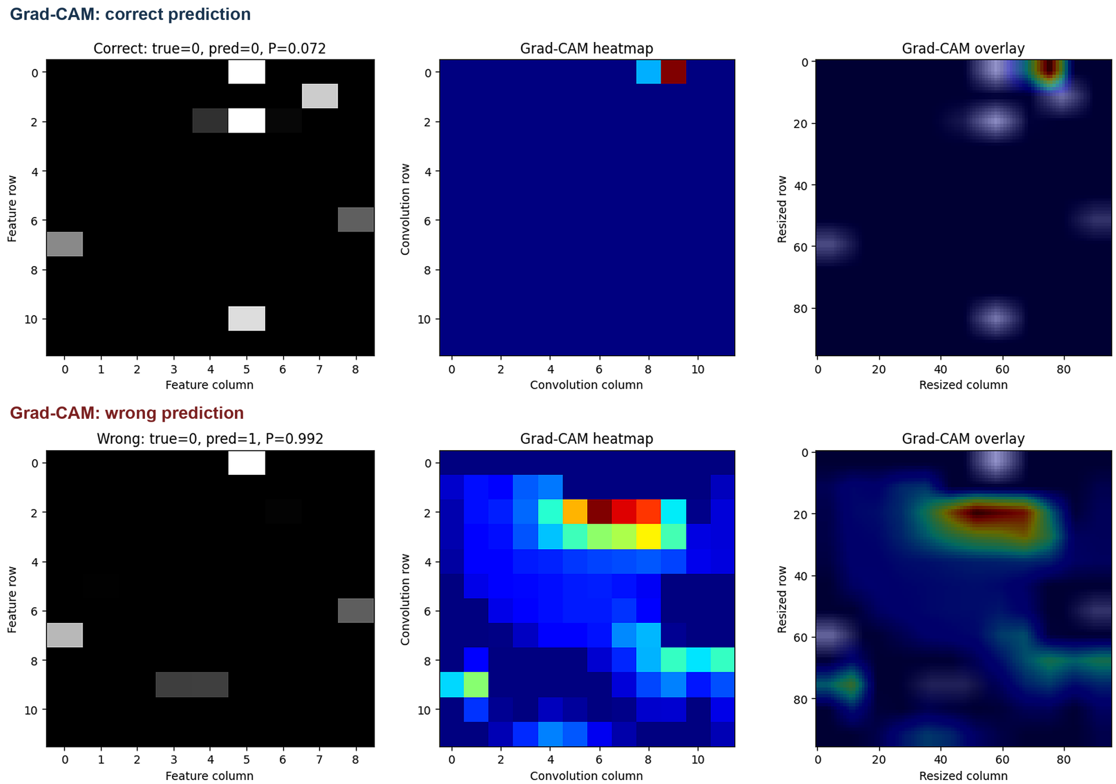

Figure 5: Grad-CAM explanations for the correctly classified benign sample (top) and the high-confidence benign false positive (bottom).

Grad-CAM weights convolutional activation maps by the gradient of the selected output [9]. Sensitivity in the correct case is concentrated around a small set of grid positions; the false positive produces a broader and stronger region. This contrast helps describe the model's internal response, but resizing and convolution mix neighboring feature slots. The heatmap cannot identify a physical region of executable code.

8.2 LIME

LIME perturbs the 108 fixed segments and fits a local surrogate around each prediction [10]. For the correct case, the largest positive weights include Maxpar (0.2083), majlv (0.1732), sig (0.1314), char (0.0990) and Data_char (0.0616). Maxpar also ranks first for the false positive, but its weight is only 0.0341; Text_char, Minpar, Infoem and Ivalss follow. The change in magnitude and supporting fields shows that the explanations are sample-specific.

Agreement between Grad-CAM and LIME can indicate that two sensitivity analyses emphasize similar slots. It does not demonstrate causality, and the imposed CSV ordering prevents the highlighted positions from being interpreted as natural malware regions.

9. What the experiments establish

The combined evidence does not support superiority of CBAM over the matched plain CNN. Validation selects CBAM without dropout, yet the untouched test set favors the plain model by 0.0217 Macro-F1 and McNemar's test shows a directional error advantage for the plain network. Five-fold means also favor the plain CNN, although the Wilcoxon result is not significant.

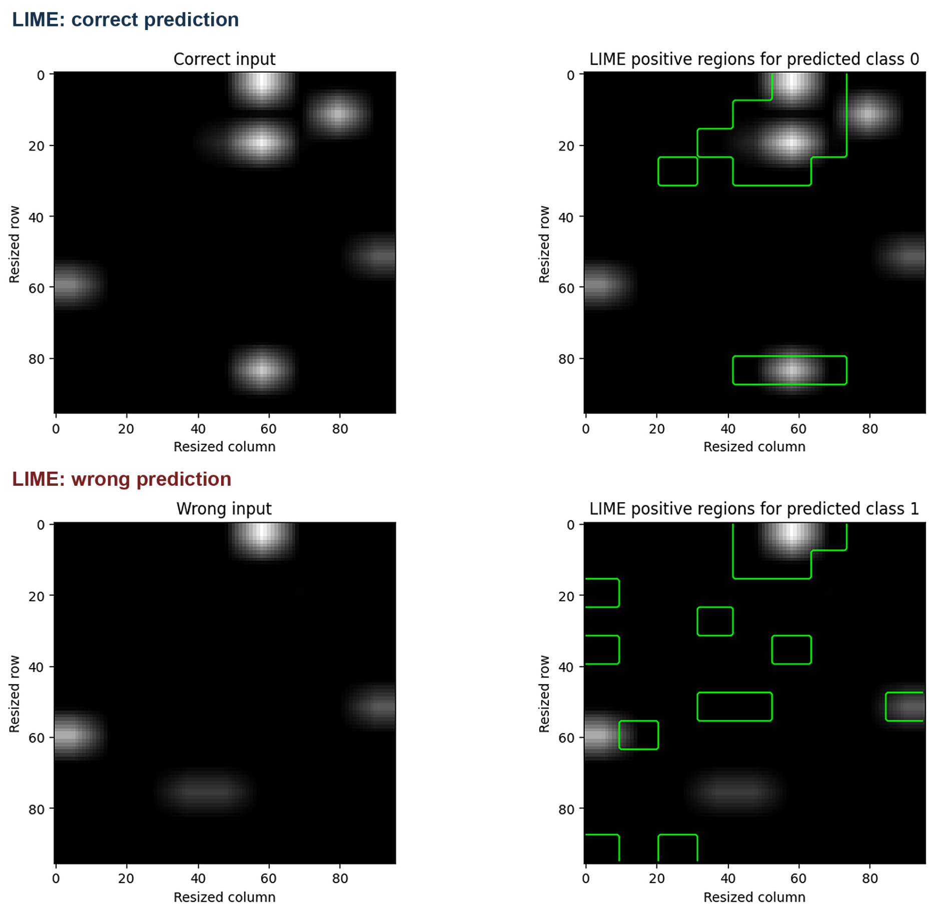

Figure 6: LIME positive regions for the correctly classified benign sample (top) and the high-confidence benign false positive (bottom).

The ablation remains informative because it separates attention from dropout, reveals the training pipeline's dependence on augmentation, and identifies how the selected attention model changes false-positive and false-negative counts. These controls prevent the validation improvement from being attributed to attention alone.

10. Limitations

Exact-vector grouping does not guarantee independence by malware family, collection source or time because those identifiers are not present. Related variants may remain across folds. The grid topology is also imposed by CSV column order; convolutional locality and spatial explanations therefore refer to an engineered arrangement rather than executable structure.

The ablation configurations were each run once, and five fold scores give the Wilcoxon test limited power. No external or temporal test set is available. Within these constraints, the defensible conclusion is negative: attention changes the error profile but is not shown to improve balanced classification.

References

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